

5-Mark Questions & Answers MPCA Unit 3: The Intel Microprocessor 8086 Architecture

Q1. Explain the internal architecture of 8086 (BIU and EU) with a neat diagram.

1. **Bus Interface Unit (BIU):** Handles external bus communication, fetches instruction code into a 6-byte FIFO queue, and computes 20-bit physical addresses using segment registers (`CS`, `DS`, `SS`, `ES`) and `IP`.

2. **E**
16-bit

Q2. Explain the programmer's model of 8086, listing all registers and their functions.

1. **Data Registers (16-bit / 8-bit split):**

- `AX`

Q3. Draw and explain the functional pin diagram of 8086 in detail.

- **40-Pin DIP Package Configuration:**

Q4. Explain memory segmentation and physical address calculation in 8086 with an example.

Physical Address Formula

Numerical Example

Q5. Explain the functioning of 8086 in minimum mode with a diagram.

Q6. Explain the functioning of 8086 in maximum mode with a diagram.

Q7. Draw and explain the timing diagram for a read operation in minimum mode.

Clock

Q8. Draw and explain the timing diagram for a write operation in minimum mode.

Clock

Q9. Explain interrupt structure and interrupt servicing in 8086.

1. **Interrupt Types:**

- **H**

